

ECE 3337: Signals & Systems Analysis

Lecture 15: Laplace Transform Applications

**Reference: Chapter 4
(Section 4.1 – 4.2)**

Please turn off your gadgets now

Math Moment: Cramer's Rule

$$4x - 3y = 11$$

$$6x + 5y = 7$$

$$\Delta = \begin{vmatrix} 4 & -3 \\ 6 & 5 \end{vmatrix} = 4 \times 5 - 6 \times (-3) = 38 \neq 0$$

$$\Delta_x = \begin{vmatrix} 11 & -3 \\ 7 & 5 \end{vmatrix} = 11 \times 5 - 7 \times (-3) = 76$$

$$\Delta_y = \begin{vmatrix} 4 & 11 \\ 6 & 7 \end{vmatrix} = 4 \times 7 - 6 \times (11) = -38$$

$$x = \frac{\Delta_x}{\Delta} = \frac{76}{38} = 2$$
$$y = \frac{\Delta_y}{\Delta} = \frac{-38}{38} = -1$$

This method will keep us organized, alert us to mistakes, and save effort



Recap: Modes of a System

Definition: The modes of an LTI system are the roots of the characteristic polynomial set to zero

- They come from the coefficients of the left-hand side of the LCCDE
- These coefficients define the system's natural behavior

Example:

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = x(t) + 2 \frac{dx(t)}{dt}$$

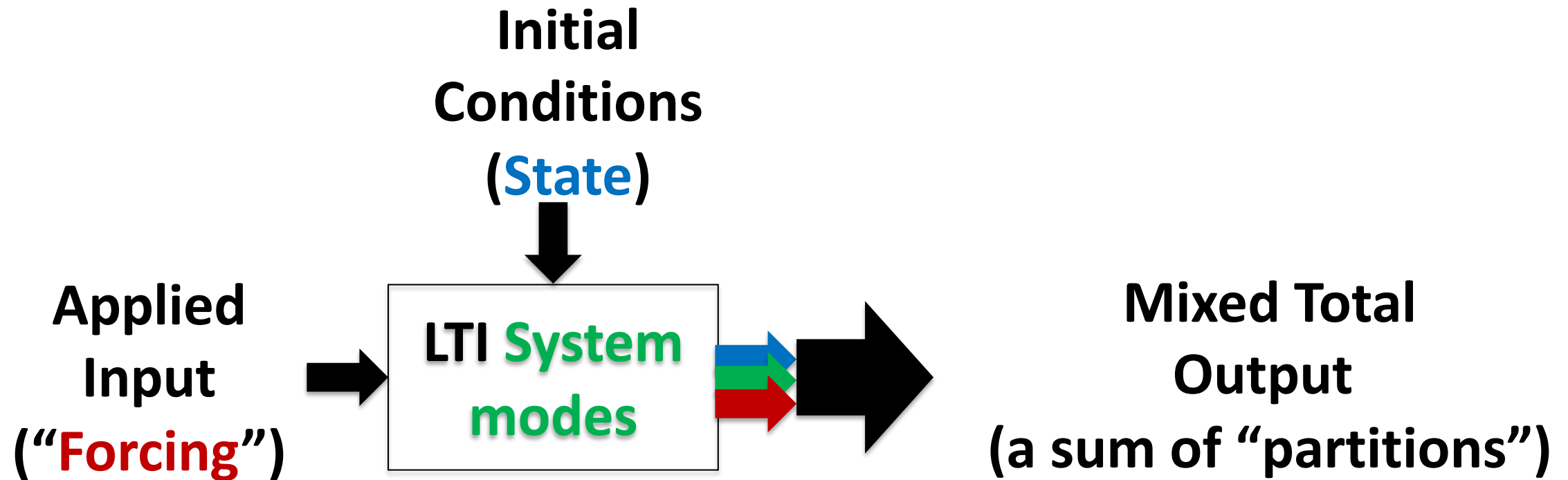
The characteristic polynomial is $s^2 + 3s + 2 = (s + 1)(s + 2)$

The modes are $\{-1, -2\}$

They manifest as poles at $\{-1, -2\}$ in the s-domain, and as exponential terms of the form e^{-t} and e^{-2t} in the time-domain



Recap: System Response Partitions



We learned to parse a system's output into its partitions:

Zero-input Response

and

Zero-state Response

Natural Response

and

Forced Response

Transient Response

and

Steady-state Response



Recap: Table of Common Convolutions

1. $u(t) * u(t) = t u(t)$

2. $e^{at} u(t) * u(t) = \left(\frac{e^{at} - 1}{a} \right) u(t)$

3. $e^{at} u(t) * e^{bt} u(t) = \left[\frac{e^{at} - e^{bt}}{a - b} \right] u(t), \quad a \neq b$

4. $e^{at} u(t) * e^{at} u(t) = t e^{at} u(t)$

5. $t e^{at} u(t) * e^{bt} u(t) = \frac{e^{bt} - e^{at} + (a - b) t e^{at}}{(a - b)^2} u(t),$
 $a \neq b$

6. $t e^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$

7. $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$



Modes of a System: Some Intuition ...

We can learn a lot about modes by looking at how the match between an input and a system mode affects the system output.

Example:

$$x(t) = e^{at}u(t)$$

$$h(t) = e^{bt}u(t)$$

$$\Rightarrow y(t) = x(t) * h(t) = \left[\frac{e^{at} - e^{bt}}{a - b} \right] u(t), a \neq b$$

Clearly, the closer a is to b , the stronger the response $y(t)$ will be. The opposite is true as well. In the extreme case when $a = b$, we get

$y(t) = te^{at}u(t)$, a potentially unbounded response.

However, if $a < 0$, the exponential falls much faster than the ramp, so the response is stable



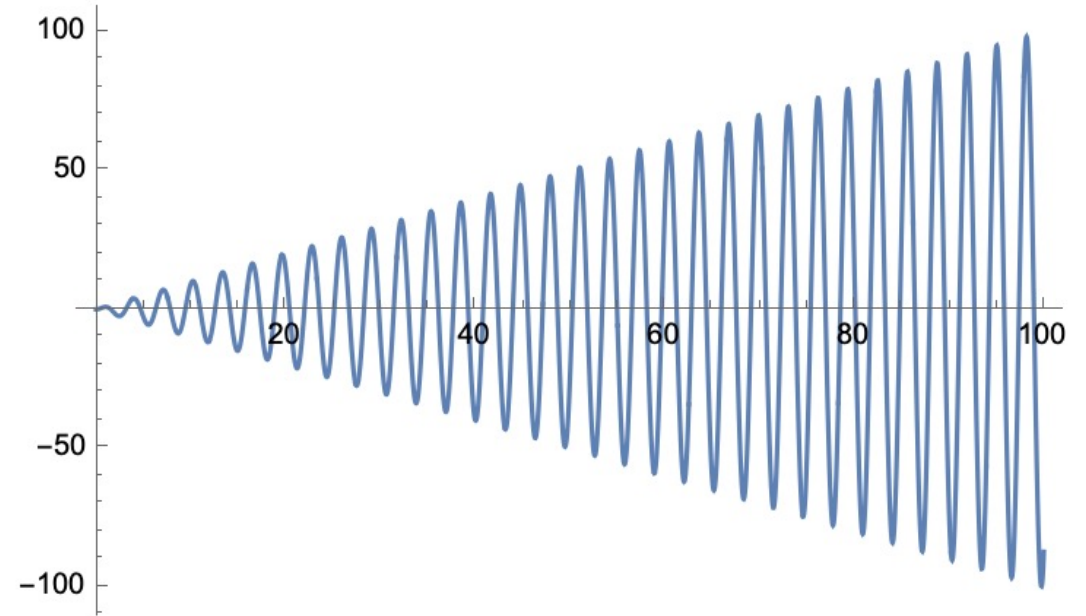
Modes of a System: Some Intuition ...

Now, let us look at the oscillatory case:

Example:

$$x(t) = e^{j\omega_1 t} u(t), h(t) = e^{j\omega_2 t} u(t)$$
$$\Rightarrow y(t) = \left[\frac{e^{j\omega_1 t} - e^{j\omega_2 t}}{j(\omega_1 - \omega_2)} \right] u(t), a \neq b$$

Clearly, the closer ω_1 is to ω_2 , the stronger the response $y(t)$ will be, and *vice versa*.



In the extreme case when

$$\omega_1 = \omega_2 = \omega,$$

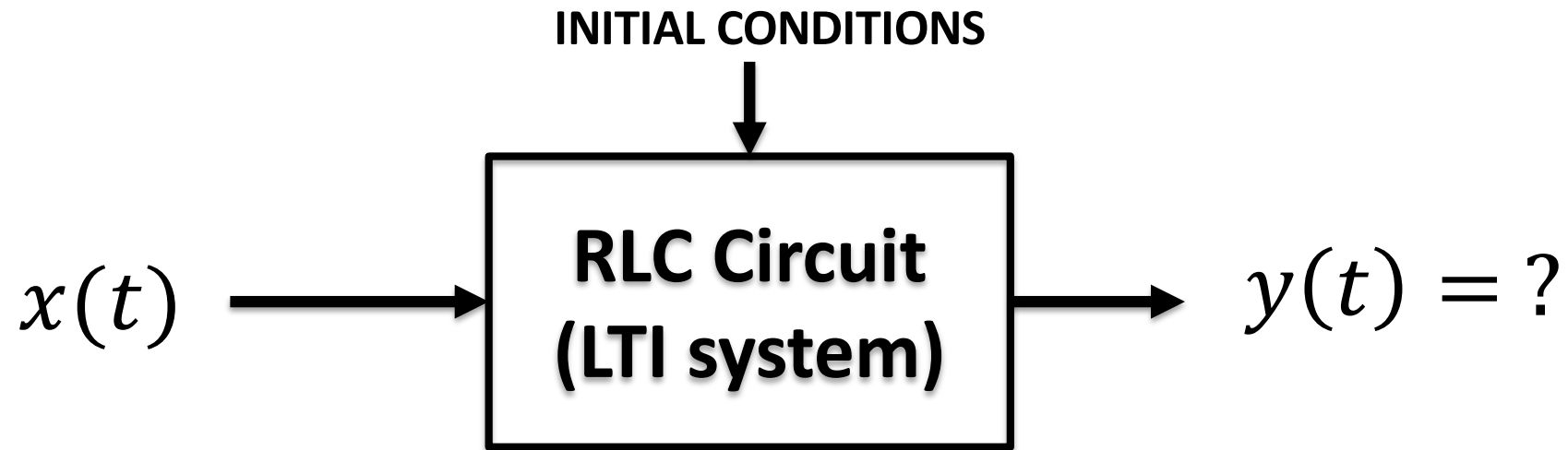
$$y(t) = t e^{j\omega t} u(t),$$

an unbounded response!

This phenomenon is known as
“resonance” (more later!)



Upping the Game



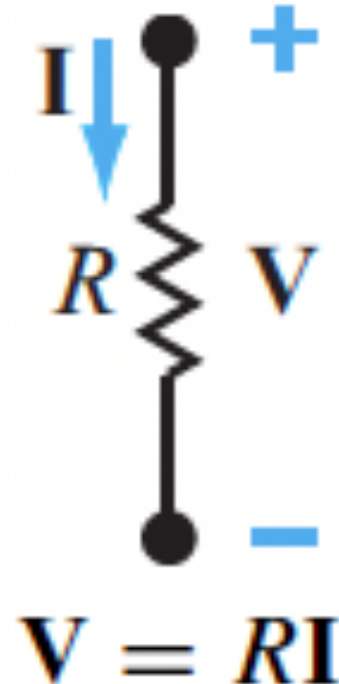
How can Laplace-domain methods help us handle complicated inputs, circuits, and initial conditions?



S-domain Representation of Resistors

A resistor does not store any energy, only capacitors and inductors can, so

- No need to account for initial conditions for a resistor



$$Z_R = R$$

IMPEDANCE
(Unit = Ohms)

NO INITIAL CONDITIONS

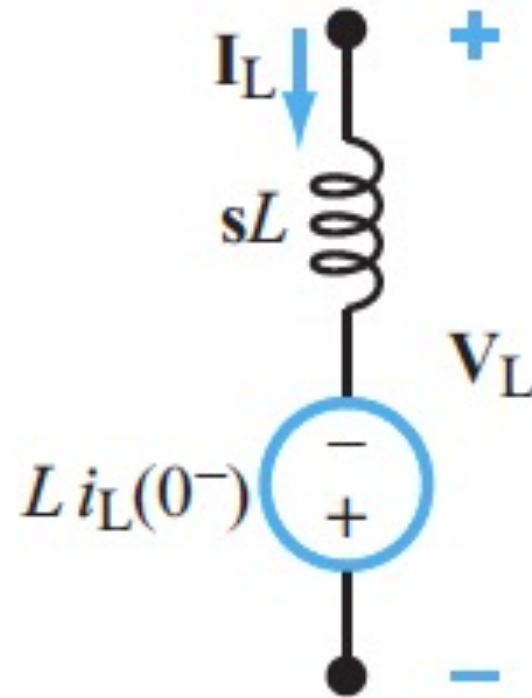
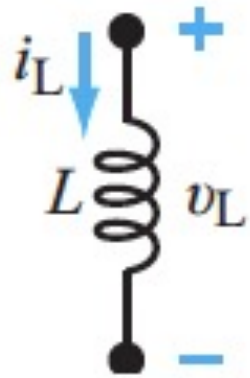
**IMPEDANCE IS
INDEPENDENT OF
FREQUENCY**



s-domain Inductor Representation

An inductor can have an initial energy store

- We need to account for initial conditions



IMPEDANCE
(Unit = Ohms)

$$Z_L = sL$$

**FREQUENCY DEPENDENT
IMPEDANCE:
BEHAVES LIKE AN OPEN CIRCUIT
AT HIGH FREQUENCIES,
& A SHORT CIRCUIT AT DC**

$$v_L = L \frac{di_L}{dt}$$

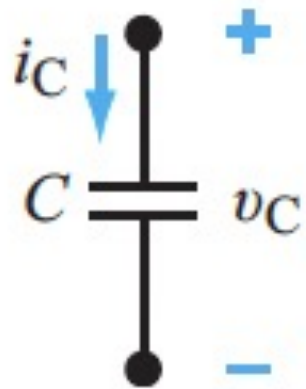
$$i_L = \frac{1}{L} \int_{0^-}^t v_L dt' + i_L(0^-)$$

$$V_L = sL I_L - L i_L(0^-)$$



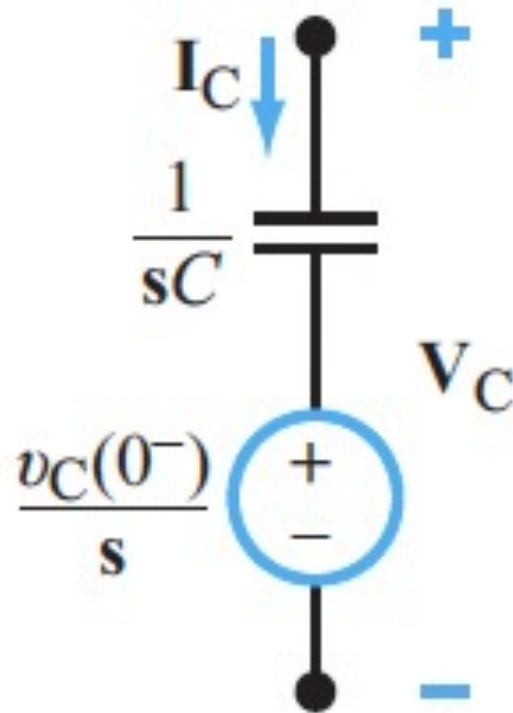
s-domain capacitor Representation

A capacitor can have an initial energy store



$$i_C = C \frac{dv_C}{dt}$$

$$v_C = \frac{1}{C} \int_{0^-}^t i_C dt + v_C(0^-)$$



$$V_C = \frac{I_C}{sC} + \frac{v_C(0^-)}{s}$$

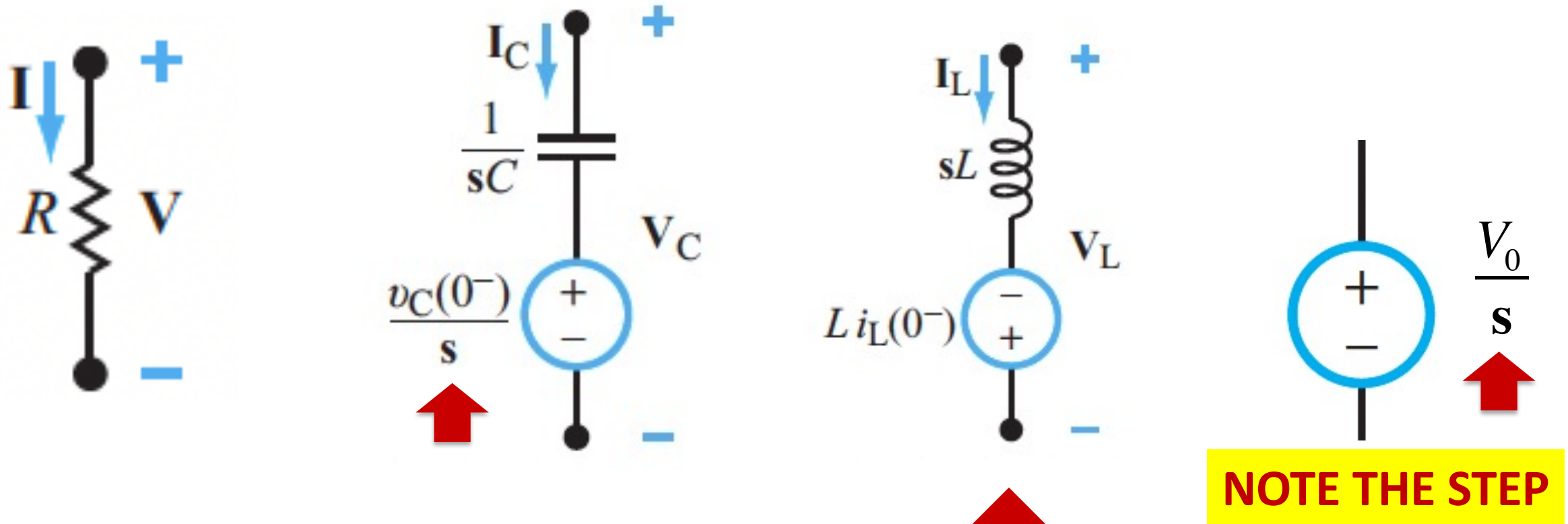
$$Z_C = \frac{1}{sC}$$

IMPEDANCE
(Unit = Ohms)

**FREQUENCY DEPENDENT
IMPEDANCE:
BEHAVES LIKE A SHORT CIRCUIT AT
HIGH FREQUENCIES AND
AN OPEN CIRCUIT AT DC**



Rewriting Circuits in Laplace Domain



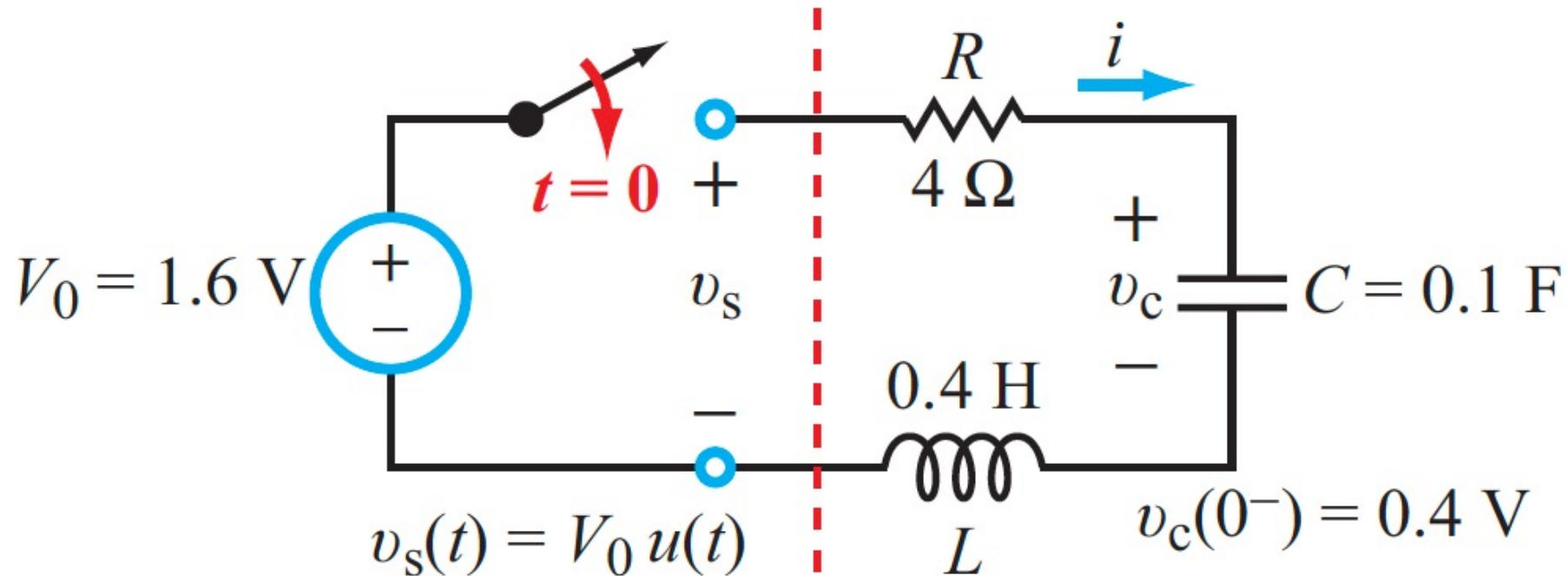
**NOTE THE BACKWARD SIGN
"REACTIVE" "KICK-BACK" VOLTAGE**



Circuit Example #1

Laplace transforms make our work easier compared to the old-fashioned method based on solving differential equations

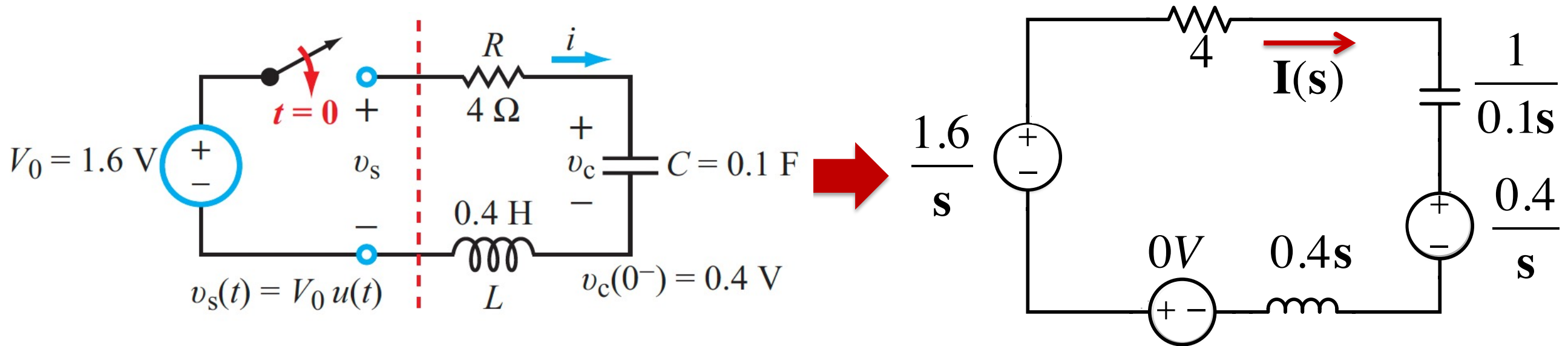
Example: Find $i(t), t \geq 0$



Initial Condition: $v_c(0^-) = 0.4 \text{ V}, \quad i(0^-) = 0 \text{ A}$



Map the Circuit to Laplace Space



We can solve for $I(s)$ directly

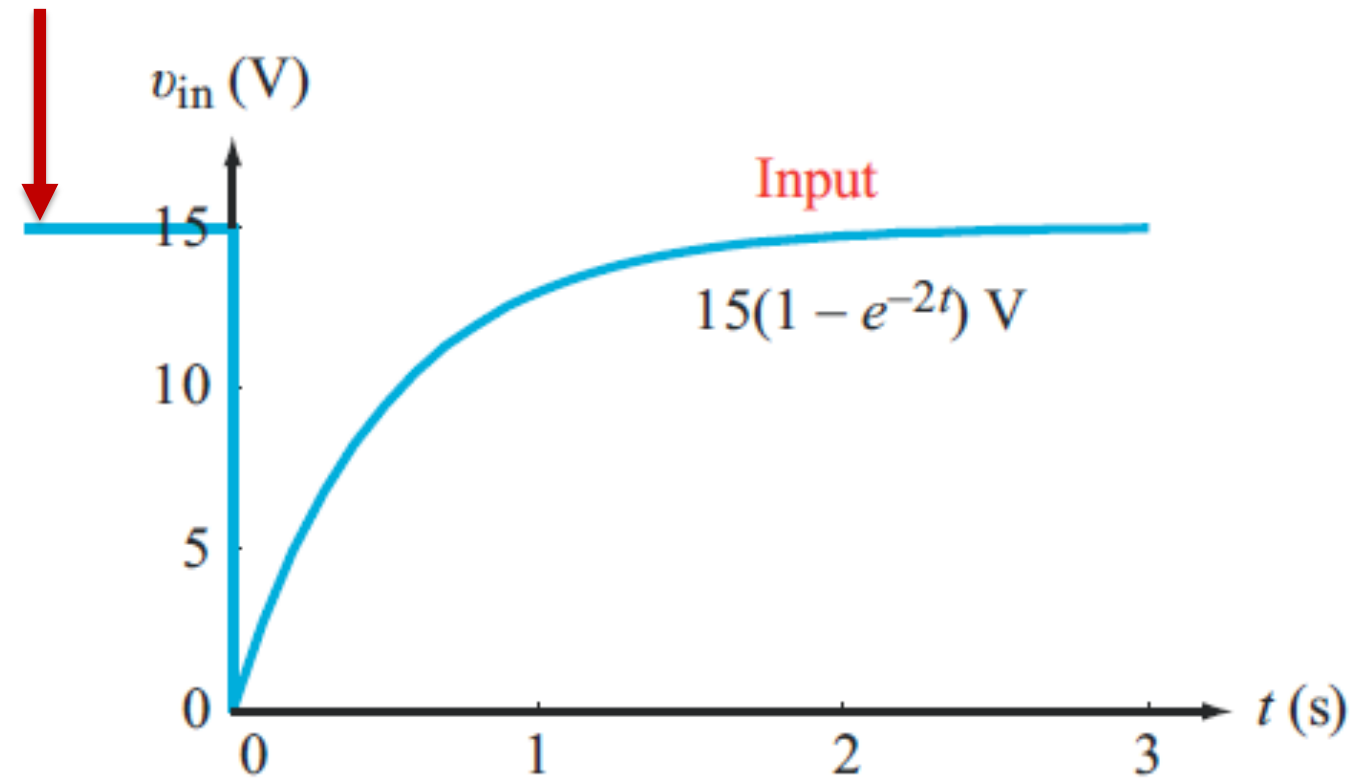
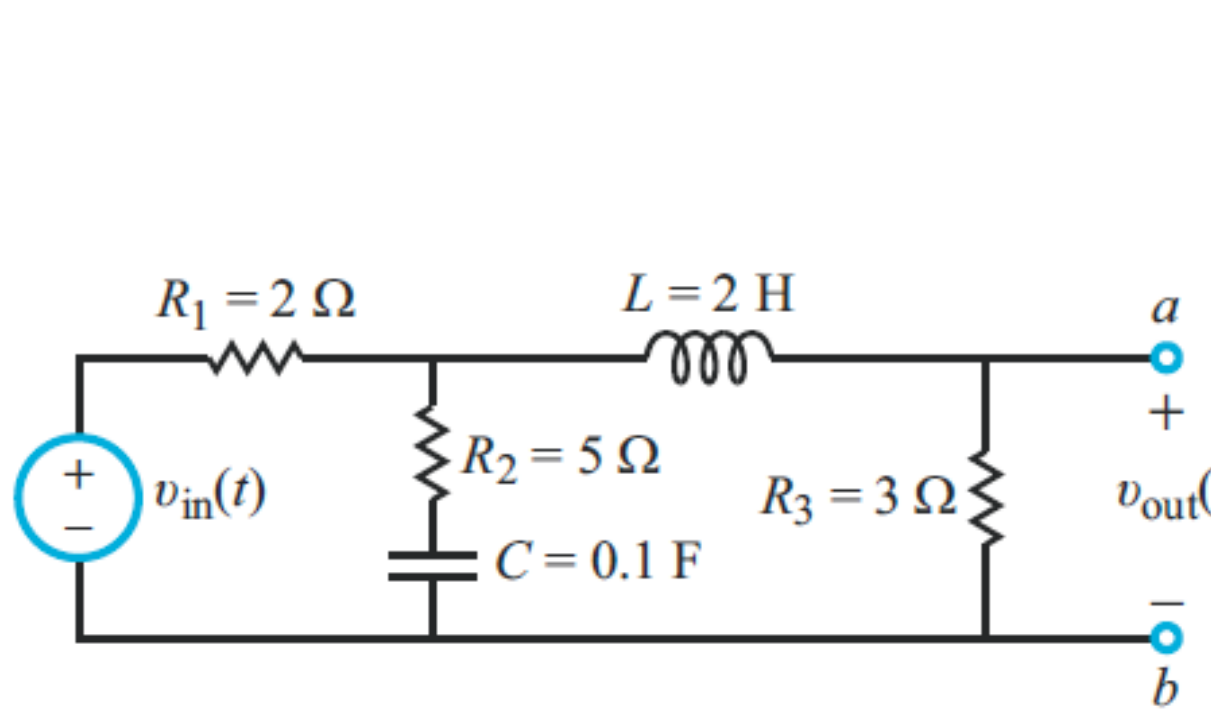
$$I(s) = \frac{\frac{1.6}{s} - \frac{0.4}{s}}{4 + \frac{10}{s} + 0.4s} = \frac{3}{(s+5)^2}$$



Example #2

Voltage was 15V for a long time

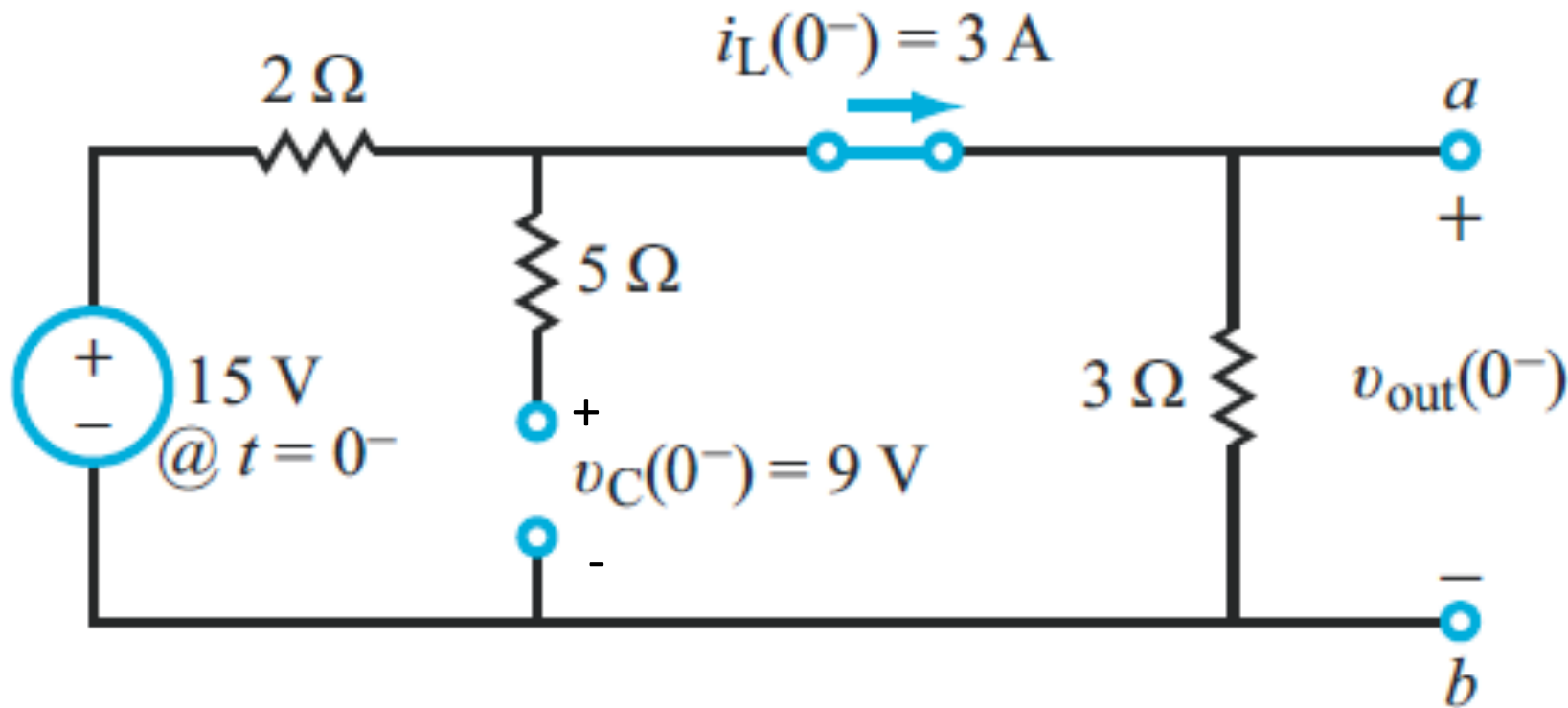
Sudden dip at $t = 0$, and then recovers to 15V



Problem: Derive an expression for $v_{out}(t)$



Step 1: Initial Condition Circuit, $t = 0^-$



Initial Conditions

$$v_C(0^-) = 9\text{ V}$$

$$i_L(0^-) = 3\text{ A}$$

$$v_{out}(0^-) = 9\text{ V}$$

Under initial steady conditions:

Inductors behave as short circuits

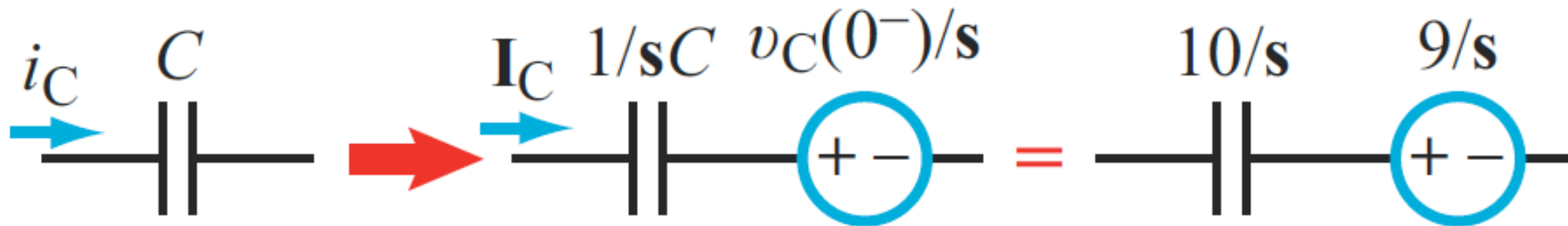
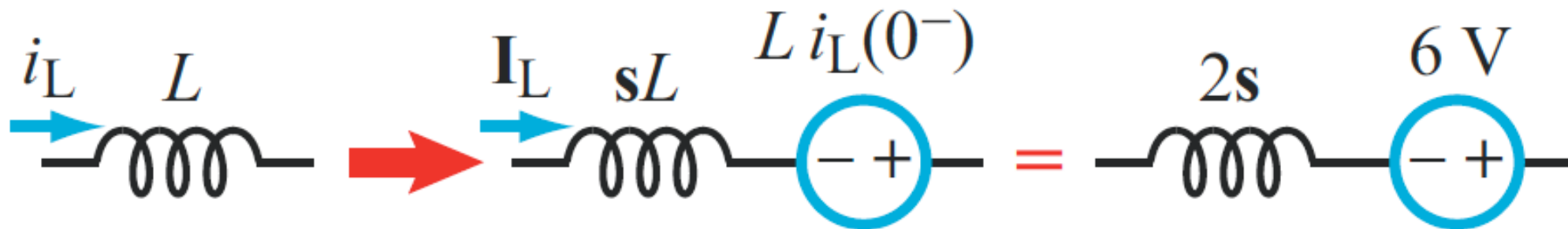
Capacitors behave as open circuits



Transform the Circuit to Laplace Space

Time Domain

s-Domain



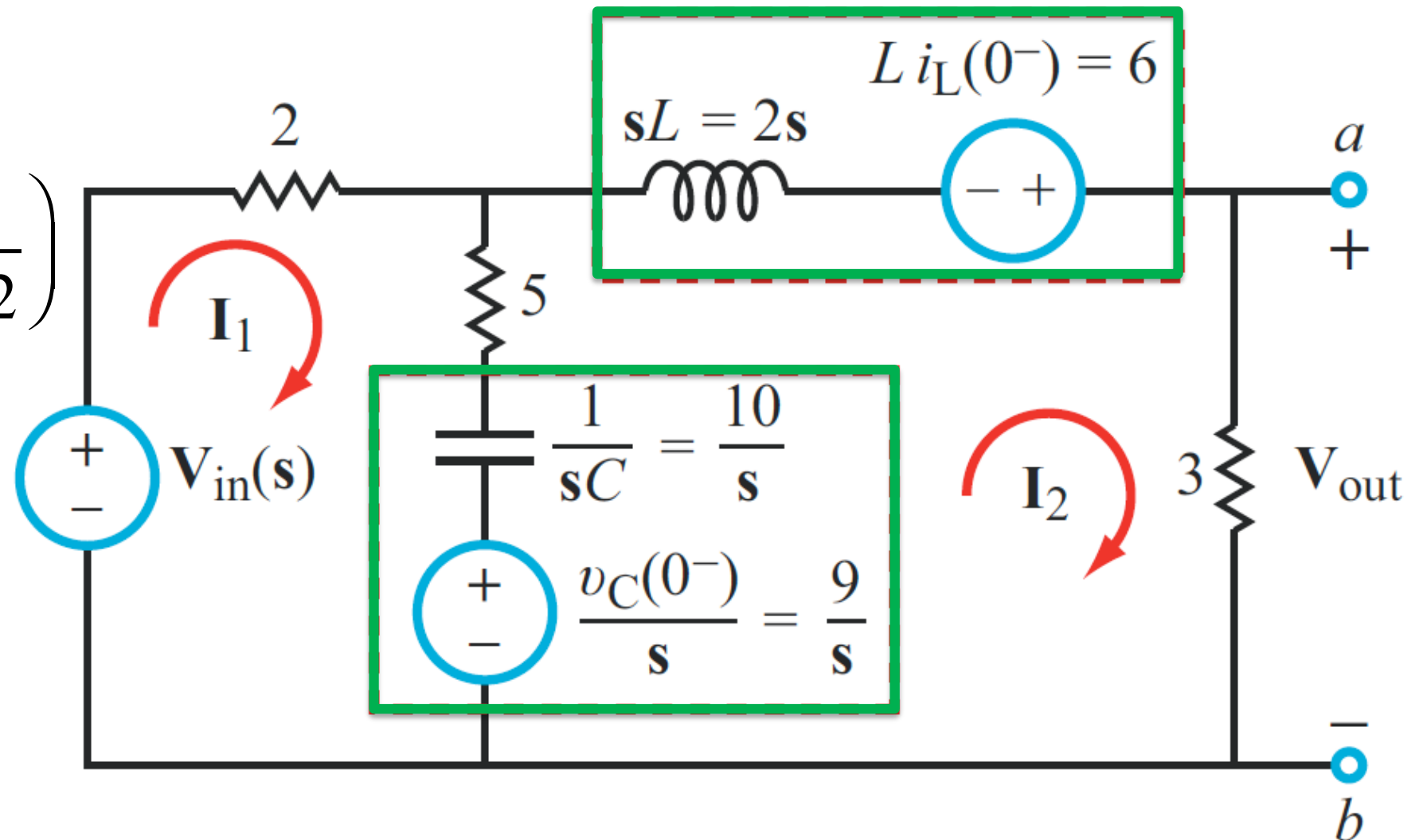


Transform Circuit to Laplace Space

$$v_{in}(t) = 15(1 - e^{-2t})u(t)$$

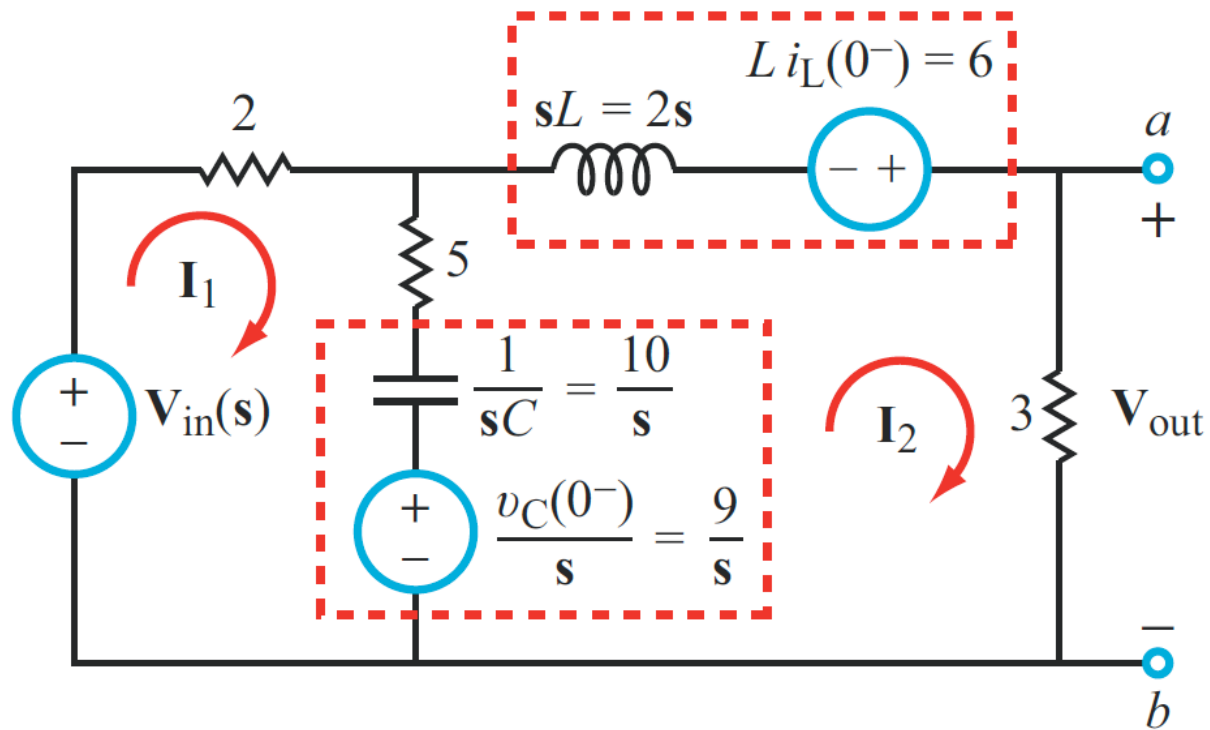
$$\Rightarrow V_{in}(s) = 15\left(\frac{1}{s} - \frac{1}{s+2}\right)$$

**THIS CIRCUIT DIAGRAM
HAS THE INITIAL
CONDITIONS "BUILT IN"**





Mesh KVL Analysis



$$\left(2 + 5 + \frac{10}{s}\right) \mathbf{I}_1 - \left(5 + \frac{10}{s}\right) \mathbf{I}_2 = V_{in} - \frac{9}{s}$$

$$-\left(5 + \frac{10}{s}\right) \mathbf{I}_1 + \left(3 + 5 + 2s + \frac{10}{s}\right) \mathbf{I}_2 = \frac{9}{s} + 6.$$



Cramer's Rule

Cramer's Rule provides us with a clean and efficient way to solve simultaneous linear equations using determinants

- It works for any number of variables, and the 2-variable case is summarized below:

$$\begin{array}{l} ax + by = e \\ cx + dy = f \end{array} \quad \Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \neq 0 \quad \text{IMPORTANT CHECK}$$

$$\Delta_x = \begin{vmatrix} e & b \\ f & d \end{vmatrix} = ed - bf, \quad \Delta_y = \begin{vmatrix} a & e \\ c & f \end{vmatrix} = af - ec$$

$$x = \frac{\Delta_x}{\Delta}, \quad y = \frac{\Delta_y}{\Delta}$$



Applying Cramer's Rule

$$\mathbf{I}_1 \left(7 + \frac{10}{s} \right) - \mathbf{I}_2 \left(5 + \frac{10}{s} \right) = \mathbf{V}_{in} - \frac{9}{s}$$

$$-\mathbf{I}_1 \left(5 + \frac{10}{s} \right) + \mathbf{I}_2 \left(8 + 2s + \frac{10}{s} \right) = \frac{9}{s} + 6$$

Multiply throughout by s : (just to simplify the algebra)

$$\mathbf{I}_1 (7s + 10) - \mathbf{I}_2 (5s + 10) = s\mathbf{V}_{in} - 9$$

$$-\mathbf{I}_1 (5s + 10) + \mathbf{I}_2 (8s + 2s^2 + 10) = 9 + 6s$$

$$\Delta = \begin{vmatrix} (7s + 10) & -(5s + 10) \\ -(5s + 10) & (8s + 2s^2 + 10) \end{vmatrix} = 14s^3 + 51s^2 + 50s$$

$$= s(14s^2 + 51s + 50)$$



Applying Cramer's Rule

$$\mathbf{I}_1(7s + 10) - \mathbf{I}_2(5s + 10) = s\mathbf{V}_{in} - 9$$

$$-\mathbf{I}_1(5s + 10) + \mathbf{I}_2(8s + 2s^2 + 10) = 9 + 6s$$

$$\Delta_{I_2} = \begin{vmatrix} (7s + 10) & s\mathbf{V}_{in} - 9 \\ -(5s + 10) & 9 + 6s \end{vmatrix} = 42s^2 + 78s + s(5s + 10)\mathbf{V}_{in}$$

Therefore:

$$\mathbf{I}_2 = \frac{\Delta_{I_2}}{\Delta} = \frac{42s^2 + 78s + s(5s + 10)\mathbf{V}_{in}}{s(14s^2 + 51s + 50)} = \frac{42s + 78 + 5(s + 2)\mathbf{V}_{in}}{14s^2 + 51s + 50}$$

$$\mathbf{V}_{in} = \frac{30}{s(s + 2)} \Rightarrow \mathbf{I}_2 = \frac{42s^2 + 78s + 150}{s(14s^2 + 51s + 50)}$$



Transform into time-domain

$$\mathbf{I}_2 = \frac{(42s^2 + 78s + 150)}{14s(s^2 + 51/14s + 50/14)}$$

Partial fraction expansion

$$\mathbf{I}_2 = \frac{3}{s} + \frac{5.32e^{-j90^\circ}}{s + 1.82 + j0.5} + \frac{5.32e^{j90^\circ}}{s + 1.82 - j0.5}$$

$$\begin{aligned} i_2(t) &= [3 + 10.64e^{-1.82t} \cos(0.5t + 90^\circ)] u(t) \\ &= [3 - 10.64e^{-1.82t} \sin 0.5t] u(t) \text{ A,} \end{aligned}$$

$$v_{out}(t) = 3i_2(t)$$

**Let's talk about
how to do this
next ...**



Big/Messy Partial Fraction Expansions

When expressions get big/messy, it makes sense to use software tools



partial fraction expansion $(42s^2+78s+150)/(14s(s^2+s(51/14)+50/14))$



Extended Keyboard

Upload

Examples

Random

Assuming "s" is a variable | Use "s^2" as a [unit](#) instead

Input:

partial fractions

$$\frac{42s^2 + 78s + 150}{14s \left(s^2 + s \times \frac{51}{14} + \frac{50}{14} \right)}$$

Result:

[Approximate form](#)

☒ [Step-by-step solution](#)

$$\frac{42s^2 + 78s + 150}{14s \left(s^2 + \frac{51s}{14} + \frac{25}{7} \right)} = \frac{3}{s} - \frac{75}{14s^2 + 51s + 50}$$



How

SimultaneousEquations.m *

```
1 % solving a system of symbolic eq
2 clear; % clear the workspace
3 syms s t i1 i2 I1 I2 eqn1 eqn2 Vi
4 Vin == 30/(s*(s+2)) % input
5 eqn1 == (7*s+10)*I1-(5*s+10)*I2 ==
6 eqn2 == -(5*s+10)*I1+(8*s+2*s^2+10
7 [I1 I2] == solve([eqn1 eqn2],[I1 I
8 i1 == ilaplace(I1) % inverse Lapla
9 i2 == ilaplace(I2) % inverse Lapla
```

Command Window

```
I2 =
(6*(7*s^2 + 13*s + 25))/(14*s^3 + 51*s^2 + 50*s)

i2 =
3 - (150*199^(1/2)*exp(-(51*t)/28)*sin((199^(1/2)*t)/28))/199

>> SimultaneousEquations

Vin =
30/(s*(s + 2))

eqn1 =
I1*(7*s + 10) - I2*(5*s + 10) == 30/(s + 2) - 9

eqn2 =
I2*(2*s^2 + 8*s + 10) - I1*(5*s + 10) == 6*s + 9

I1 =
(3*(4*s^3 + 39*s^2 + 102*s + 100))/((s + 2)*(14*s^3 + 51*s^2 + 50*s))

I2 =
(6*(7*s^2 + 13*s + 25))/(14*s^3 + 51*s^2 + 50*s)

i1 =
(75*exp(-(51*t)/28)*(cos((199^(1/2)*t)/28) - (9*199^(1/2)*sin((199^(1/2)*t)/28))/199)/14 - (15*exp(-2*t))/2 + 3

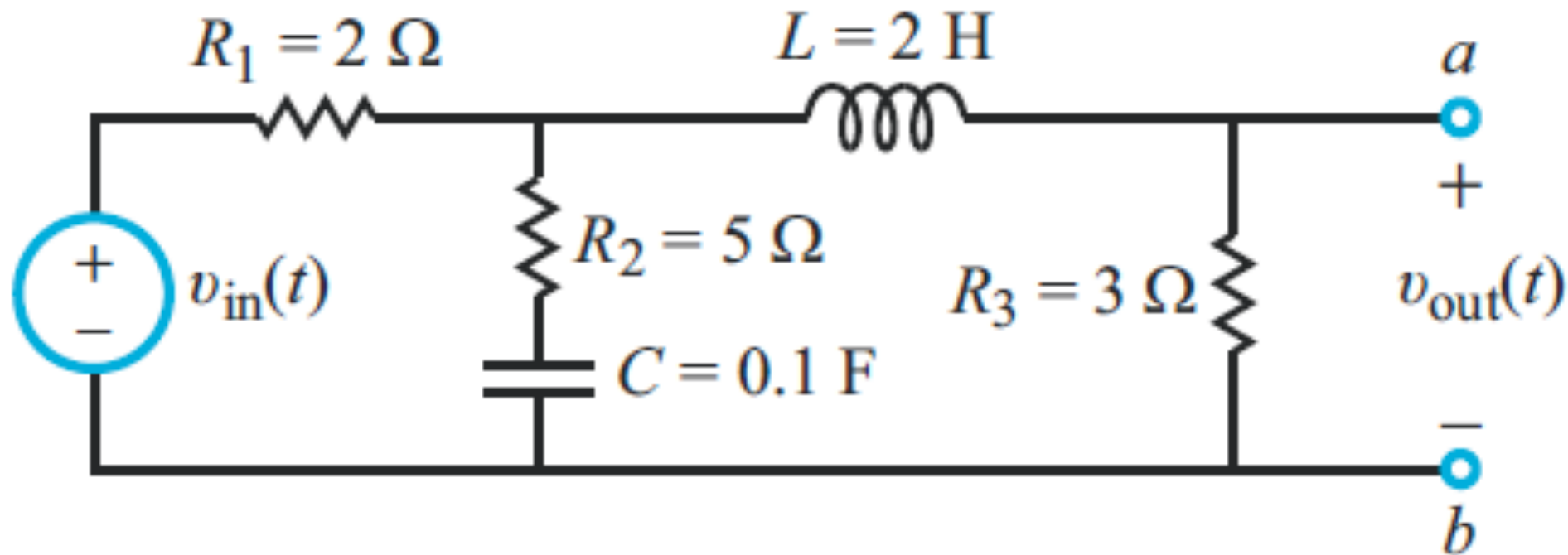
i2 =
3 - (150*199^(1/2)*exp(-(51*t)/28)*sin((199^(1/2)*t)/28))/199

fx >>
```



Let's Change the Input

Let's consider the same circuit as in the previous example, but with a changed input:



NEW INPUT: $v_{in}(t) = \delta(t)$



Let's Change the Input

Earlier, we had calculated:

$$\mathbf{I}_2 = \frac{\Delta_{I_2}}{\Delta} = \frac{42s^2 + 78s + s(5s + 10)\mathbf{V}_{in}}{s(14s^2 + 51s + 50)} = \frac{42s + 78 + 5(s + 2)\mathbf{V}_{in}}{14s^2 + 51s + 50}$$

NEW INPUT: $v_{in}(t) = \delta(t) \Rightarrow \mathbf{V}_{in}(s) = 1$

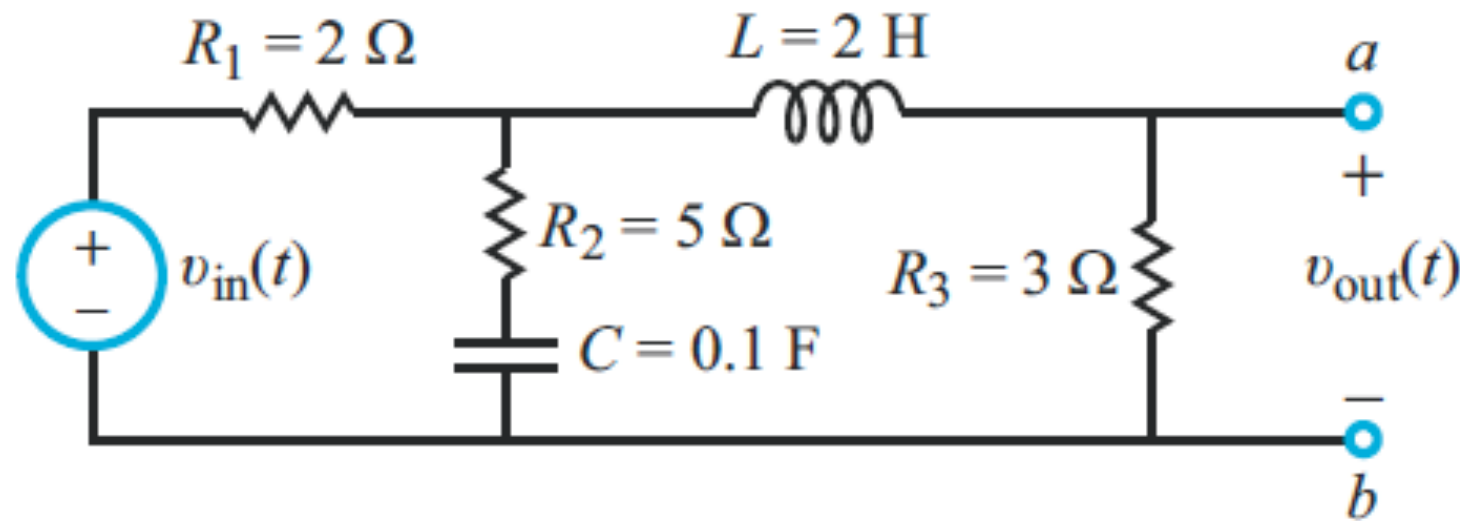
WHAT ARE THE MODES?

The roots of the denominator are $1.82 \pm j0.5$



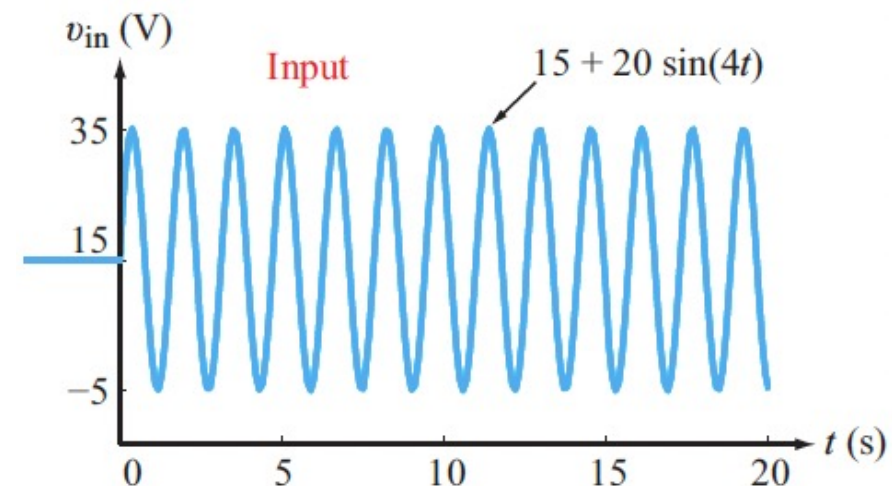
Let's Change the Input

Let's consider the same circuit as in the previous example, but with a changed input:



NEW INPUT:

$$v_{in}(t) = \begin{cases} 15\text{V} & t < 0 \\ [15 + 20 \sin 4t]u(t) & t \geq 0 \end{cases}$$





Example (cont.)

$$v_{in}(t) = \begin{cases} 15V & t < 0 \\ [15 + 20 \sin 4t]u(t) & t \geq 0 \end{cases}$$

$$V_{in}(s) = \frac{15}{s} + \frac{20\omega}{s^2 + \omega^2} = \frac{15}{s} + \frac{80}{s^2 + 16}$$

The circuit can be solved again using this changed input.

For details, follow Example 4-2 of the textbook

Let us focus on understanding the output



Interpreting the Output

STEADY-STATE DC Output

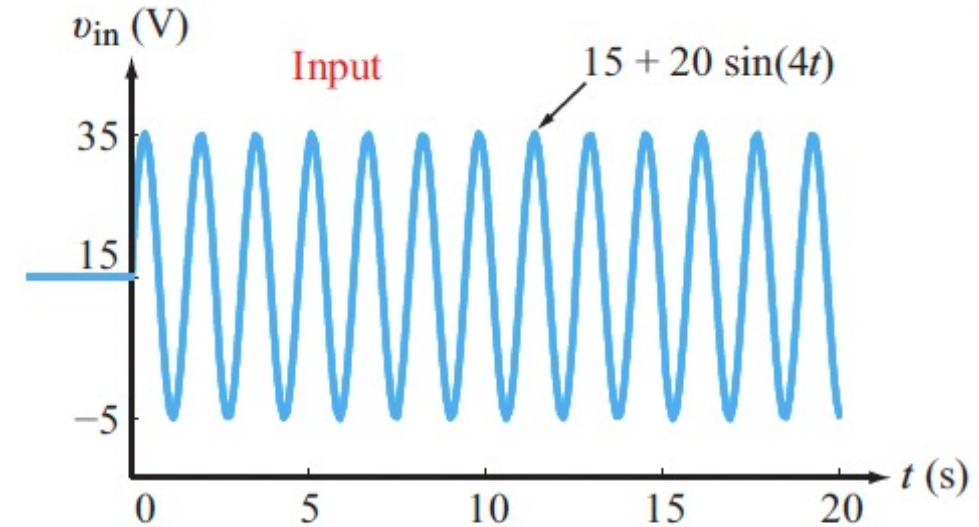


$$v_{out}(t) = [9 + 5 \cos(4t - 157^\circ)]$$

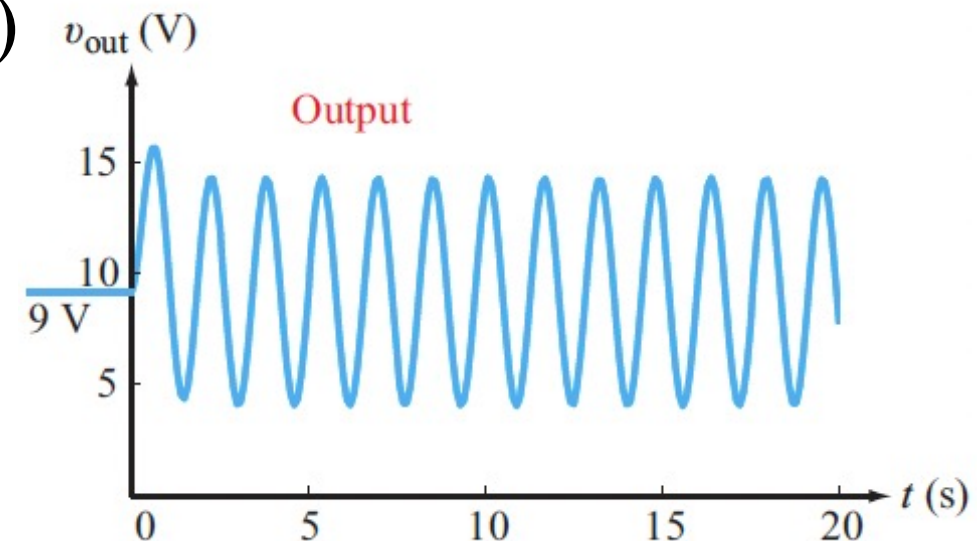
STEADY-STATE OSCILLATORY
OUTPUT (same frequency as the input)

$$+ 4.73e^{-1.82t} \cos(0.5t - 14^\circ)]u(t)$$

OSCILLATORY TRANSIENT
Can we relate this to the modes
 $\{1.82 \pm j0.5\}$?

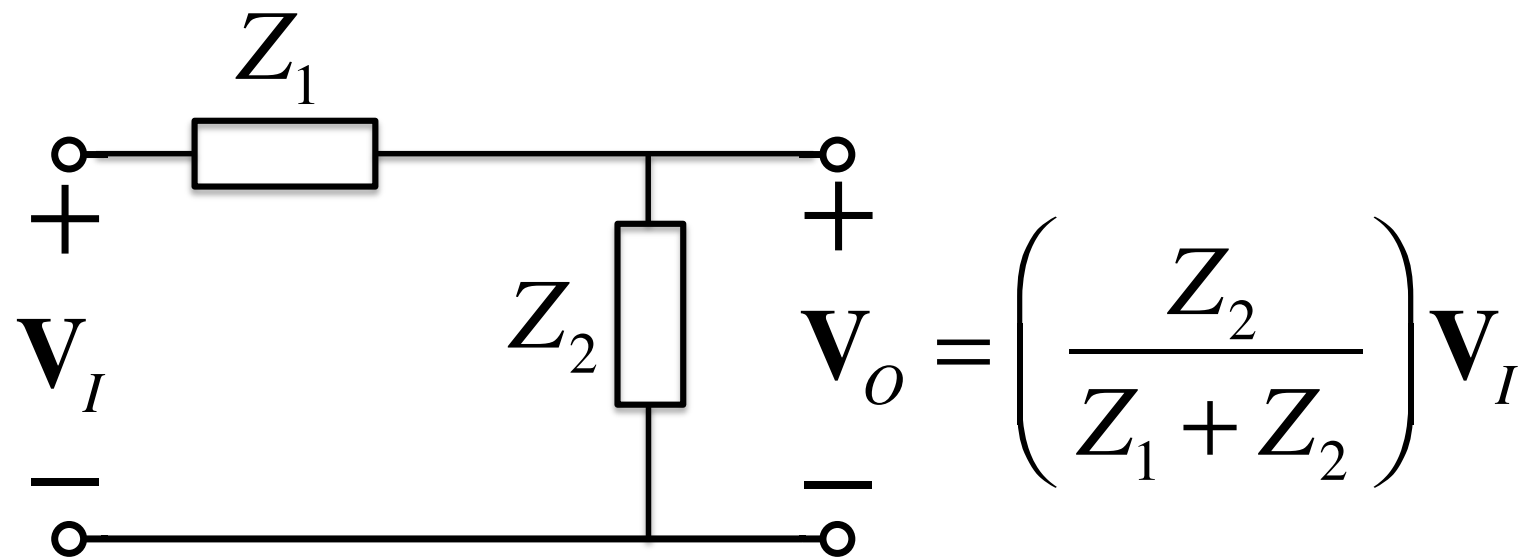


(a) Waveform of $v_{in}(t)$





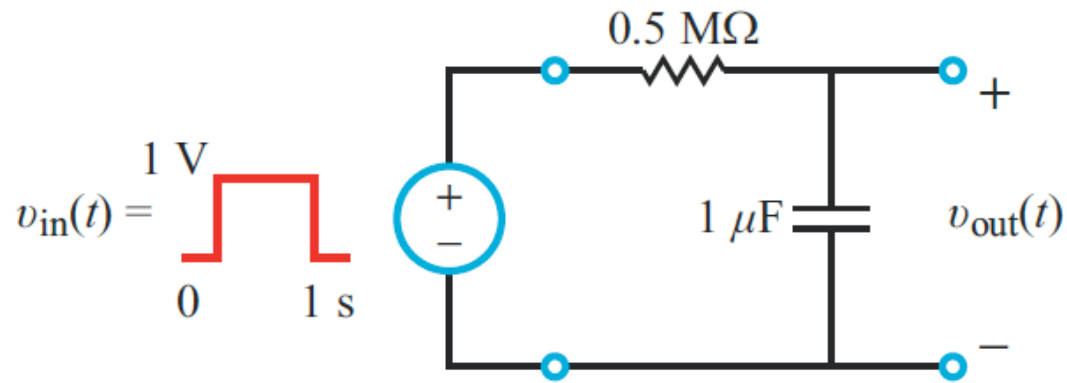
Recap: Voltage Divider



NOTE: FORMULAS FOR IMPEDANCES IN SERIES AND PARALLEL ARE SIMILAR TO THOSE FOR RESISTORS



RC Circuit Example

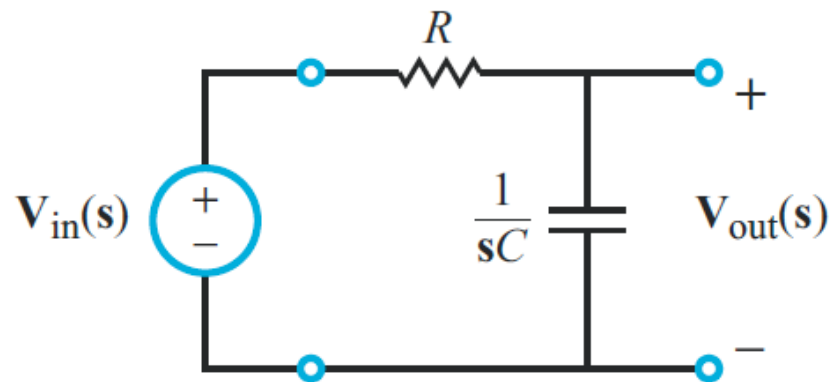


(a) RC lowpass filter

$$v_{in}(t) = u(t) - u(t - 1)$$

$$\Rightarrow V_{in}(s) = \frac{1}{s} - \frac{1}{s} e^{-s}$$

DELAGER



(b) s-domain

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1/sC}{R + 1/sC} = \frac{2}{s + 2}$$

VOLTAGE DIVIDER

$$V_{out}(s) = \left(\frac{1}{s} - \frac{1}{s} e^{-s} \right) \frac{2}{(s + 2)} = \frac{2}{s(s + 2)} - e^{-s} \frac{2}{s(s + 2)}$$



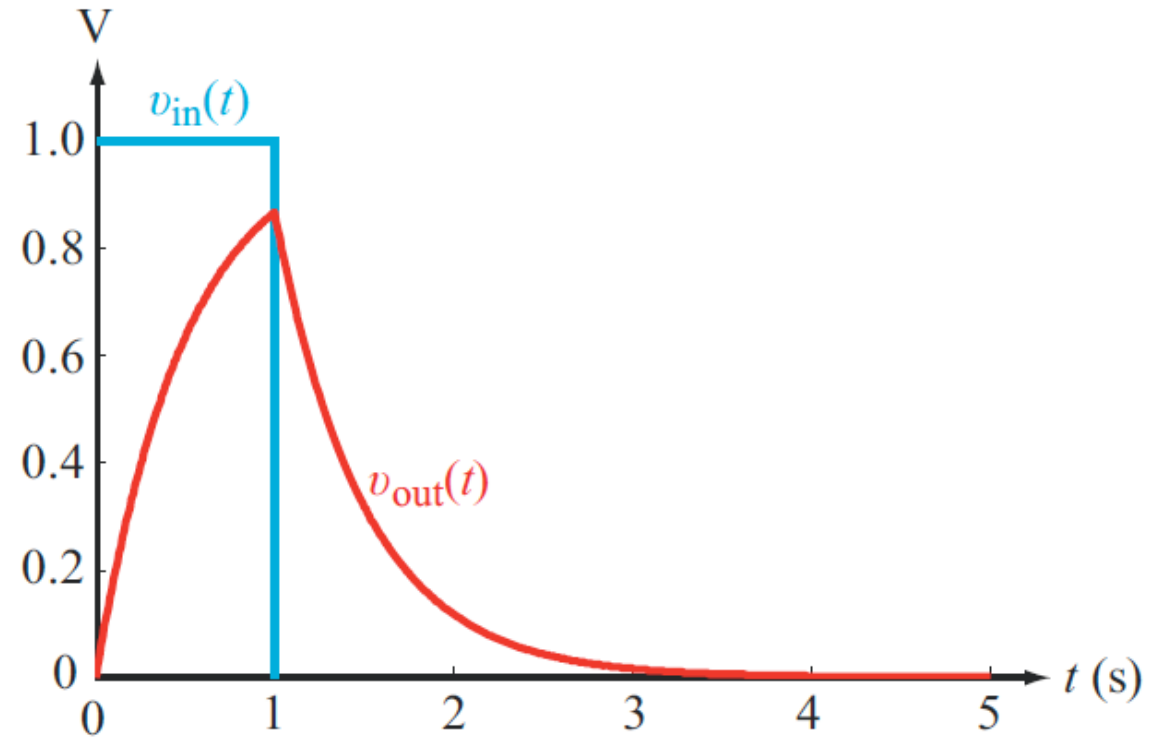
RC Circuit Example

PARTIAL FRACTION EXPANSION

$$\frac{2}{s(s+2)} = \frac{1}{s} - \frac{1}{(s+2)}$$

$$\mathbf{V}_{out}(s) = \frac{1}{s} - \frac{1}{(s+2)} - \frac{1}{s}e^{-s} + \frac{1}{(s+2)}e^{-s}$$

$$\Rightarrow v_{out}(t) = [1 - e^{-2t}]u(t) - [1 - e^{-2(t-1)}]u(t-1)]$$



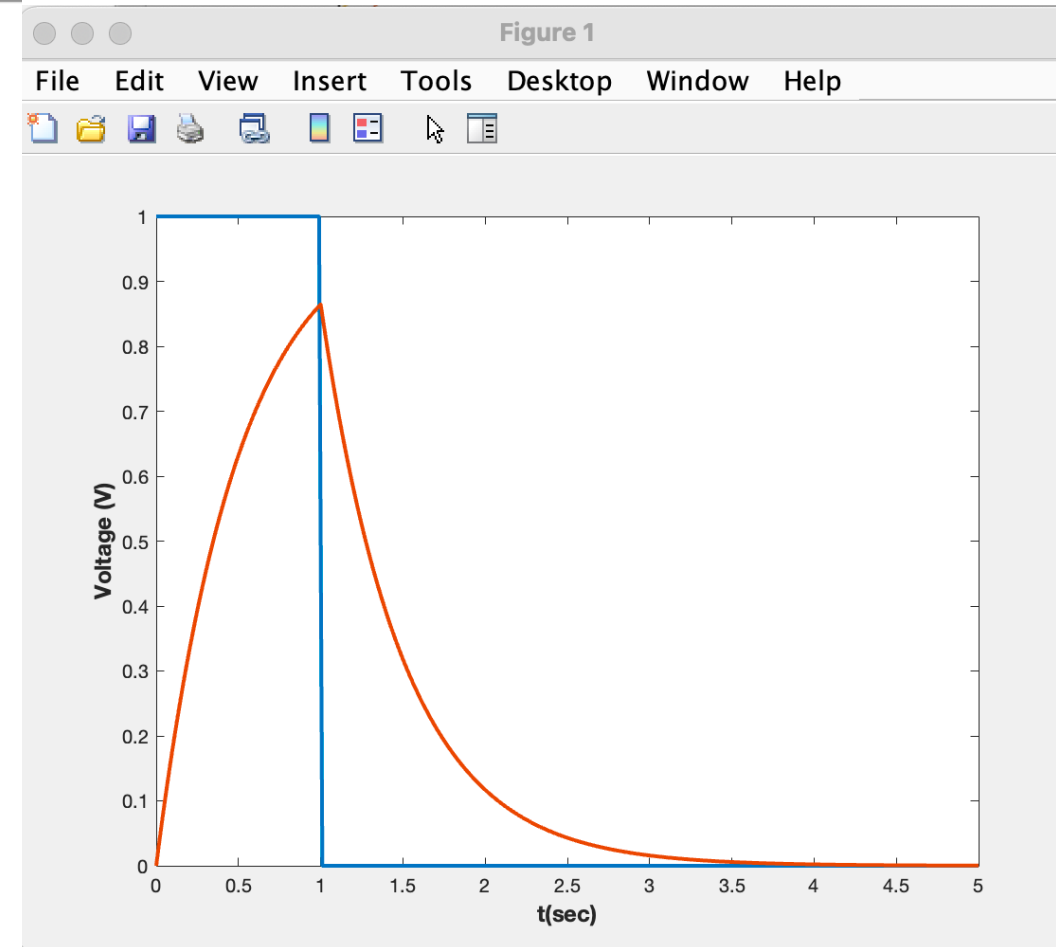
(c) Output response

**MUCH EASIER
THAN BEFORE**



How to do this in MATLAB

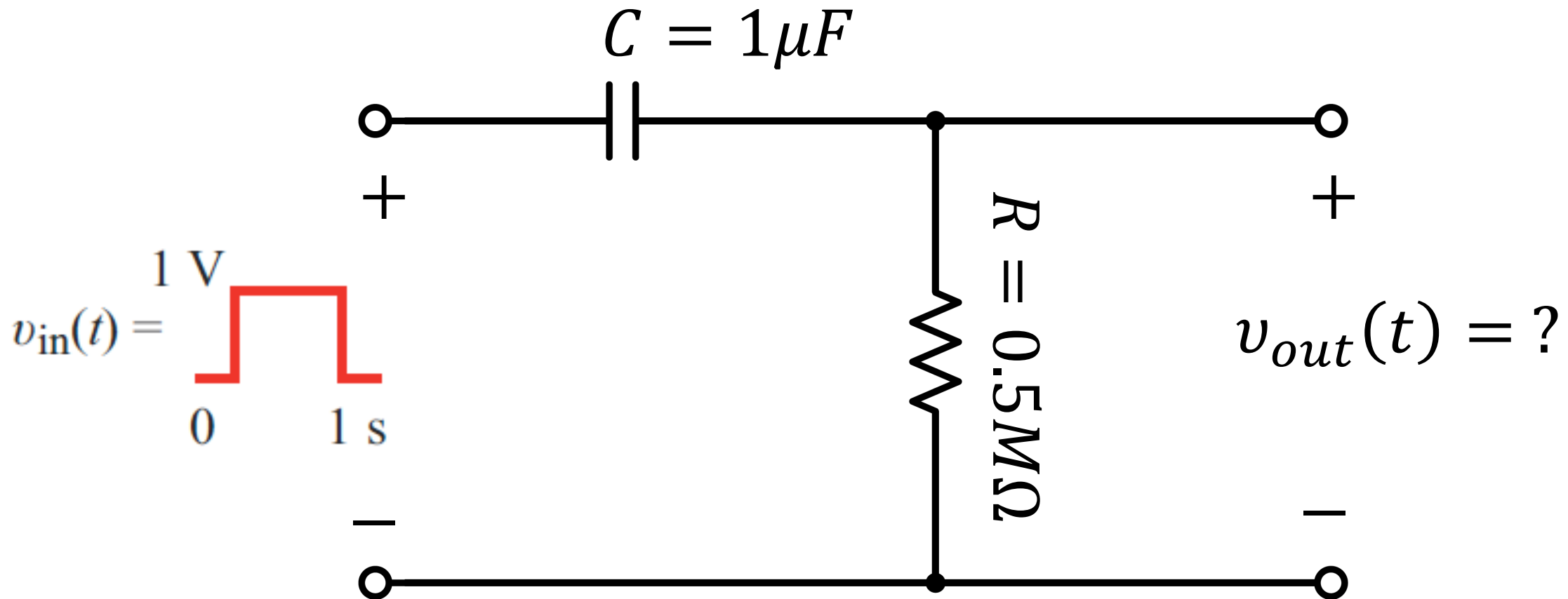
```
RCPulseExample.m * +
1 % This example needs the Symbolic Math Toolbox of MATLAB
2 clear; % Start by clearing the workspace
3 syms t s X(s) H(s) Y(s) x(t) h(t) y(t); % Declare the symbols
4
5 X(s) = 1/s-exp(-s)/s; % specify the input X(s)
6 H(s) = 2/(s+2); % specify the system H(s)
7 Y(s) = X(s)*H(s); % System output in the Laplace domain
8 x(t) = ilaplace(X); % inverse Laplace transform x(t)
9 y(t) = ilaplace(Y); % inverse Laplace transform y(t)
10
11 % Plot the results
12 time = 0:0.01:5; % array of time points for plotting
13 figure; plot(time,x(time),time,y(time),'LineWidth',2);
14 xlabel('t(sec)','FontSize',12,'FontWeight','bold');
15 ylabel('Voltage (V)','FontSize',12,'FontWeight','bold');
16
17 % Pause for the user and close the figure
18 pause;
19 close all;
```





Activity

Repeat the RC circuit example, but with the resistor and capacitor swapped, using the Laplace-domain method. The initial conditions are zero.





Summary

The Laplace transform allows us to analyze complicated circuits with initial conditions and various inputs, more easily

Step 1: Transform the circuit to the s -domain, paying attention to the initial conditions for capacitors and inductors

Step 2: Transform the input to the s -domain

Step 3: Use KCL and KVL in the s -domain to solve for the output

Step 4: Inverse transform to the time domain, using partial fraction expansion

This is a very general method – it applies to all kinds of circuits. When the math gets messy, we can use symbolic math tools.



For the Next Class

Work through the textbook examples in Section 4-1 and 4-2, and watch the YouTube video lecture on op amp circuits:

- <https://youtu.be/8QfigKEPz4A>



Study Sections 4-5 & 4-6 (skip 4-3 and 4-4)

How to Study:

- 1. Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts**
- 2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
- 3. Write down your questions and bring them to class**